



"Your Partner in Innovative Refrigeration and Advanced Testing Solutions."

CLIMATIC TEST CHAMBER THERMAL CYCLIC TEST CHAMBER



Sri Easwari Scientific Solution Pvt Ltd.

+91 94444 27748, +91 94444 59430

info@sess.co.in www.sess.co.in

SRI EASWARI SCIENTIFIC SOLUTION PVT LTD

Sri Easwari Scientific Solution Pvt Ltd (SESS) chennai is a leading name in the design and manufacturing of environmental test chambers. We are proud to be trusted by prestigious institutions and organizations such as ISRO, DRDO, IITs, and NITs. Our high-quality, innovative testing solutions are engineered to meet the rigorous demands of various sectors, ensuring precision, reliability, and efficiency.

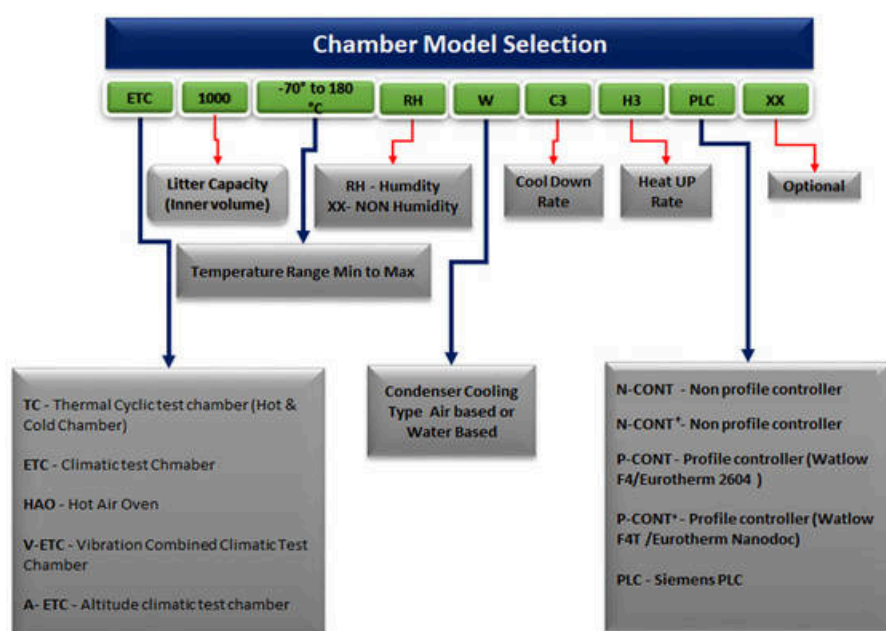


Our Key Products

- Climatic Test Chambers
- Thermal Shock Test Chambers
- Salt Spray Test Chambers
- Rapid Temperature Cycling Chambers
- Burn-In Test Chambers
- Solar Radiation Test Chambers

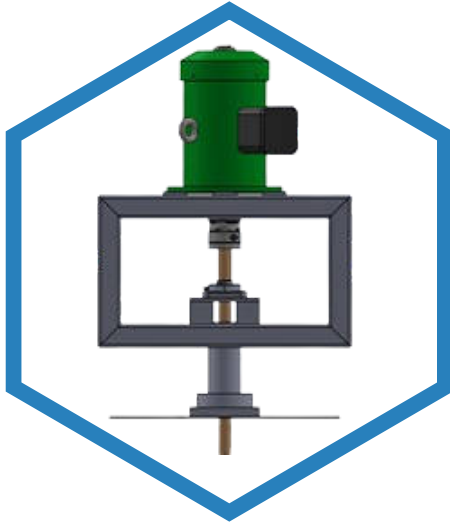


THERMAL CYCLIC TEST CHAMBER / CLIMATIC TEST CHAMBER ETC - XXXX/XXX TO XXX /XX/X/CX/HX/XXX/XX



Stand Model litter capacity
Thermal cyclic test chamber (TC),
Climatic test Chamber (ETC),
Altitude - Thermal cyclic test chamber (ATC),
Altitude - Climatic test chamber (AETC),
All dimensions in mm

Litter	W	D	H
22	300	250	300
30	350	250	350
64	400	400	400
100	500	400	600
180	580	450	750
250	600	535	750
340	600	800	700
380	600	800	800
500	800	650	950
600	850	730	900
750	800	800	950
800	1100	800	950
1000	1000	1000	1000
1200	1000	1130	1020
1500	1000	1500	1000
1800	1100	1600	1020
2000	1000	2000	1020
2250	1100	2000	1020



In our climatic test chamber's air circulation system, we incorporate the following components and features:

- 1. Fan/Blower:** The chamber is equipped with a fan or blower featuring a continuous-duty rated motor. This motor efficiently circulates air within the chamber.
 - 2. Conditioning Space:** Inside the main chamber, there's a dedicated conditioning area with baffles, housing essential components like heaters and cooling coils. These elements play a crucial role in air conditioning.
 - 3. Uniform Temperature:** Conditioned air is introduced into the main chamber to ensure consistent temperature distribution throughout the workspace, maintaining a stable temperature environment during tests.
 - 4. Laminar Airflow:** The chamber is designed for laminar airflow, achieved using a fan motor and impeller with an appropriate capacity. This setup guarantees even distribution of temperature and humidity within the chamber, adhering to specified limits.
 - 5. Motor Shaft:** TC/ETC - The motor shaft used for chamber air circulation is constructed with SS 304 without any joints. The motor is positioned outside the chamber, while an extended-length SS 304 shaft drives the fans. This design ensures smooth and reliable operation while preserving the chamber's internal environment.
- ATC/AETC** - "The mechanical design of the chamber's air circulation motor is constructed with a sturdy shaft connected by a flexible coupling. The inner shaft is made of SS 304 material and is connected to the blower drum, which features a double vacuum seal mechanism. The motor is positioned outside the chamber, and it drives the blower drum through the inner SS 304 shaft. This design ensures smooth and reliable operation while maintaining the integrity of the chamber's internal environment.
- 6. Motor Winding:** For high-temperature chambers, we utilize motor windings of H class, whereas for chambers operating below 130°C, F class motor windings are employed.

These features combine to create an efficient and precise air circulation system within the climatic test chamber, facilitating accurate and dependable test conditions for various experiments and applications.

INCONEL-TYPE AIR HEATERS WITH PTFE WIRE AND SILICON END CAP



Heating System:

The chamber is equipped with sealed Inconel tube heaters to maintain a uniform temperature. These heaters are meticulously designed to ensure even heat distribution throughout the chamber. Heat input is regulated by solid-state relays, controlled through a PID controller programmer.

One notable advantage of Inconel tube heaters is their exceptional ability to withstand high temperatures and corrosive environments. Inconel, known for its outstanding durability and heat resistance, is well-suited for prolonged use in challenging conditions.

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Compared to stainless steel heaters, Inconel heaters exhibit superior resistance to oxidation, corrosion, and high-temperature degradation. They are engineered to deliver reliable and efficient heat transfer without direct radiation onto the test item, ensuring a uniform heating process that doesn't adversely affect the test specimen.

In conclusion, the use of sealed Inconel tube heaters is highly recommended for maintaining precise and uniform chamber temperatures while ensuring the long-term performance and durability of the heating system.

Refrigeration System Based on Model Selected:

For single-stage configurations, the system features either air or water-cooled refrigeration, incorporating a Hermetic compressor from Embraco, Danfoss, or Emerson, or a semi-hermetic compressor from Bitzer, Dorin, or Bock. These compressors operate with the environmentally friendly refrigerant R404A.



The system includes essential safety features such as HP/LP cut-off switches, air inlet condensers/temperature protectors, backup fuse protection, expansion valves, and is engineered for continuous and trouble-free operation.

For double-stage cascade configurations, the system comprises either air or water-cooled refrigeration, equipped with a Hermetic compressor from Embraco or Danfoss, or a semi-hermetic compressor from Bitzer, Dorin, Bock, or Emerson. These compressors utilize environmentally friendly refrigerants R404A and R23. Safety features such as HP/LP cut-off switches, air inlet condensers/temperature protectors, backup fuse protection, expansion valves are included, ensuring continuous and trouble-free operation.



Our chamber's refrigeration system is designed for optimal performance, with variations in specifications based on different models.

Single-stage compressor for temperatures up to -40°C , and a double-stage compressor for temperatures below -70°C . We select our compressors from reputable brands like Bitzer, Dorin, Bock, Danfoss, SECOP, or Embraco.

Environmentally friendly refrigerants are used, with R404A for single-compressor systems and R404A/R23 for cascade systems.

Safety features include HP/LP cut-off switches, pressure transmitters for continuous monitoring, and suction line temperature sensors for enhanced safety.

The system can have a water or air-cooled condenser based on the model. Temperature monitoring and safety limits are integrated into our system for effective control and protection.

The compressor and condenser unit are positioned below the chamber for optimal performance.

Control Panel & Wiring:

A separate control panel will be provided, attached to the main chamber, which will house the programmer, on/off switches, fuses, contactors, indicating lamps, etc. Channel type wiring will be implemented using suitable copper wires with appropriate current ratings, along with marking ferrules, crimped dowel terminals, elmex connectors, and so on.

The electrical and instrumentation panel will be meticulously wired in compliance with international standards to ensure safety and reliability. Each electrical cable will be tagged with ID numbers, and the corresponding details will be clearly documented in the maintenance manual. Comprehensive circuit diagrams and relevant information will also be included.



To facilitate accurate and efficient design, we utilize SolidWorks 3D software for developing our drawings. This advanced software enables us to create detailed structural and electrical drawings that adhere to industry standards and specifications. By utilizing SolidWorks, we can ensure precise and comprehensive documentation of the electrical components and wiring layout of the Thermal cyclic test chamber.

Our dedication to utilizing modern software tools and obtaining relevant certifications reflects our unwavering commitment to delivering superior products that meet the highest industry standards and exceed customer expectations.

Solid state relays for control for dry heater.

OPTION 1: N-CONT - NON PROFILE CONTROLLER (SUITABLE FOR ETC, NON PROFILE CONTROL SERIES)

Highlights

- Universal Inputs (RTD/mA/V for Temperature & %RH) with Selection for Dry/Wet Configuration
- Independent Self Tune PID or On-Off Control Loops for Temperature & %RH
- Compressor Control Output with Time Delay
- Programmable Alarms & Retransmission Outputs for Temperature & %RH



Features

- 24V or 12V or 5V DC Excitation Voltage for Transmitters
- Relay or SSR Drive Outputs for Heating, Humidification & Compressor Control
- DC Volts / Current Retransmission Outputs
- Standby Mode for Use as Indicator with Alarms
- Optional RS485 MODBUS/RTU Serial Communication Port
- Universal Supply Voltage : 85~264 VAC, 50/60 Hz
- DIN Standard Dimensions (mm) : 96(H) X 96(W) X 100(D)
- Relay Output for Alarm

N-CONT⁺ - Non profile controller (Suitable for TC, Non profile Control Series) 3216 PID Temperature Controller

Specification

- Input Type : TC, RTD, mV, mA, CT
- PV Accuracy : <0.25%
- IP Rating : IP65, NEMA12
- Display Type : Main: 4 digits Lower: 5 character starburst (3216/08/04) 9 character starburst (32h8)
- Control Types : On/Off, PID, VP
- Alarm Types : Hi, Lo, Dev, Sensor break, Event, Heater fail
- Supply Voltage : 24V dc/ac 85-264V ac
- SP Programmer : 4 Ramp + 4 Dwell



OPTION 2 - PROFILE CONTROLLER (SUITABLE FOR ETC & TC PROFILE CONTROL SERIES)

P-CONT SERIES

Eurotherm Make Model 2604 Advanced Process Controller / Programmer. The 2604 is a 1/4 DIN plug-in high stability temperature and process controller designed to be used in demanding applications. It has a dual five digit display and a two line LCD panel for alarm, status and user defined messages.

Specification

- High stability, precision and functionality
- One, two or three control loops
- PID, valve, ratio, cascade and override control
- Set point programming
- High speed ModBus®, DeviceNet™ and Profibus® communications
- Test chambers, autoclaves, kilns, furnaces, fore hearths
- Melt Pressure, Carbon Potential



Watlow F4 - The SERIES F4 ¼ DIN industrial ramping temperature controller meets the requirements of the most demanding ramp soak controller processing applications. Easy to set up and operate, the ramp soak controller's programming features and proven performance capabilities are ideally suited for environmental chamber or furnace and oven applications. Single and dual channel versions are available.

Competitively-priced, the SERIES F4 ramping temperature controller features a four line, high-definition LCD interface display for quick and easy profile programming and controller configuration. Its 16-bit microprocessor ensures accuracy and delivers performance advantages you can count on from a Watlow controller.



Features

- Guided 100 step, 20 profile ramp and soak programmable memory supports a wide range of processing applications
- High-definition, four line LCD controller interface display simplifies setup and operation
- Menu customization for enhanced process monitoring
- High-performance, 16-bit microprocessor provides precise process control
- Application versatility with universal inputs

P-CONT⁺ SERIES

Eurotherm Make Nanodac - Chart Recorder

The nanodacTM recorder/controller offers the ultimate in graphical recording combined with PID control for a box of its size. This secure data recording device with accurate control is enhanced by a full colour, VGA display to bring a crystal clear operator interface to even the smallest of machines.



The 3.5" TFT display offers incredibly clear visualization of process parameters with a wide selection of configurable views to best suit the application. The views include: Horizontal and vertical trends, Horizontal and vertical bar graphs, Numeric, Alarm panel, Alarm status, and control loops. The unit also provides user wiring from the front of the product for detailed configuration without the need to connect to a PC.

Watlow F4T with INTUITION® The F4T with INTUITION® temperature process controller offers a wide range of field removable I/O modules for maximum design flexibility. Configurations can be custom tailored to meet the scaling needs of a tremendous range of equipment and applications while providing exactly the hardware types required for compatibility. The F4T controller also features a 4.3 inch, color, graphical touch panel.

Combining power, flexibility and functionality, this new controller offers unmatched versatility, and its best-in-class ease of use could very well make user manuals a thing of the past.



TEST CHAMBER ALARM

In our all chamber indicators are available on the soft screen of the HMI itself, along with a buzzer option for the following:

- SPP trip
- Air Circulation Fan motor OLP
- Over Temperature in Chamber inner
- HS Compressor OLR
- HS Compressor over temperature
- HS Compressor HP trip
- HS Compressor LP trip
- LS Compressor OLR
- LS Compressor over temperature
- LS Compressor HP trip
- LS Compressor LP trip
- During transfer of basket (missing of the basket)
- ETC model - Water level Low
- Air Heater protection etc....



CERTIFICATION



THANK YOU!

Presented By
SESS TECHNICAL TEAM



Sri Easwari Scientific Solution Pvt Ltd.

No. 2/298, ANE Garden, Perumal Kovil Street, Paraniputhur Post,
Iyyappanthangal, Chennai - 600 122.

 +91 94444 27748, +91 94444 59430

 info@sess.co.in

 www.sess.co.in

    @sesschennai